Solutions — Percents, Lines, Quadratics, Pythagoras

Part A — Percentage Word Problems

1. Jacket price.

> **Solution:** Discounted price $= 80(1-0.25) = 80(0.75) = 60$. With tax: $60(1+0.06) = 60(1.06) = \$63.60$.

2. Percent increase (population).

> **Solution:** Increase $= 13440 - 12000 = 1440$. Percent $= \dfrac{1440}{12000} \cdot 100\% = 12\%$.

3. Original price from sale price.

> **Solution:** Sale $= 0.85(\text{original}) \Rightarrow 714 = 0.85P \Rightarrow P = \dfrac{714}{0.85} = 840$ dollars.

4. Markup percent.

> **Solution:** $50(1 + x/100) = 59 \Rightarrow 1 + x/100 = 1.18 \Rightarrow x = 18\%$.

5. Score growth and repeat.

> **Solution:** Percent increase $= \dfrac{90-72}{72} = \dfrac{18}{72} = 25\%$. Apply again: $90(1.25) = 112.5$.

6. Salt amount.

> **Solution:** $0.08 \cdot 350 = 28$ mL of salt.

Part B — Slope, Intercepts, and Graphing Lines

7. Slope from two points.

Solution: $m = \dfrac{-5-7}{2-(-4)} = \dfrac{-12}{6} = -2$.

8. Equation with slope and a point.

Solution: Point—slope: $y - 2 = -3(x-4) \Rightarrow y = -3x + 12 + 2 = -3x + 14$.

9. Intercepts of $3x - 2y = 12$.

Solution: x-int: set $y = 0 \Rightarrow 3x = 12 \Rightarrow x = 4 \Rightarrow (4, 0)$.
y-int: set $x = 0 \Rightarrow -2y = 12 \Rightarrow y = -6 \Rightarrow (0, -6)$.
Slope—intercept: $-2y = -3x + 12 \Rightarrow y = \frac{3}{2}x - 6$.

10. Line through a point and y-intercept.

Solution: Slope $m = \dfrac{5-(-4)}{6-0} = \dfrac{9}{6} = \dfrac{3}{2}$. Equation: $y = \dfrac{3}{2}x - 4$.

11. Compare L_1 and L_2.

Solution: Slopes: 2 and $-1/2$. Product $= 2\cdot(-1/2) = -1$ so lines are perpendicular (and different b so they intersect).

12. Slope with parameters.

Solution: $m = \dfrac{11-3}{(a+4)-a} = \dfrac{8}{4} = 2$. Using point $(a, 3)$: $y - 3 = 2(x-a) \Rightarrow y = 2x - 2a + 3$.

13. Describe graphing $y = -\dfrac{3}{2}x + 6$.

Solution: y-int $(0, 6)$. x-int: set $y = 0 \Rightarrow 0 = -\frac{3}{2}x + 6 \Rightarrow x = 4$. Another point: move right 2, down 3: from $(0, 6)$ to $(2, 3)$. Plot $(0, 6), (4, 0), (2, 3)$ and draw the line.

Part C — Simple Quadratic Polynomials

14. Factor $x^2 - 11x + 24$.

Solution: $(x - 3)(x - 8)$.

15. Solve $x^2 + 5x + 6 = 0$.

Solution: $(x + 2)(x + 3) = 0 \Rightarrow x = -2, -3$. Intercepts $(-2, 0)$ and $(-3, 0)$.

16. Vertex and y-intercept of $y = x^2 - 6x + 5$.

Solution: Vertex $x = -\frac{b}{2a} = \frac{6}{2} = 3$, $y(3) = 9 - 18 + 5 = -4$. Vertex $(3, -4)$. y-int: at $x = 0$, $y = 5$.

17. Evaluate $f(x) = 2x^2 - 3x - 5$.

Solution: $f(-2) = 2(4) - 3(-2) - 5 = 8 + 6 - 5 = 9$. $f(4) = 2(16) - 12 - 5 = 32 - 17 = 15$.

18. Consecutive integers.

Solution: Let $n(n + 1) = 72 \Rightarrow n^2 + n - 72 = 0 \Rightarrow (n + 9)(n - 8) = 0 \Rightarrow n = -9$ or 8. Integers: $(-9, -8)$ or $(8, 9)$.

Part D — Pythagoras and Distance

19. Hypotenuse $7, 24$.

Solution: $c = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$.

20. Ladder.

Solution: Length $= \sqrt{6^2 + 12^2} = \sqrt{36 + 144} = \sqrt{180} = 6\sqrt{5} \approx 13.42$ ft.

21. Distance $A(-3, 5)$ to $B(9, -7)$.

Solution: $\sqrt{(9 - (-3))^2 + (-7 - 5)^2} = \sqrt{12^2 + (-12)^2} = \sqrt{144 + 144} = \sqrt{288} = 12\sqrt{2}$.

22. Rectangle sides from diagonal.

Solution: $x^2 + (x + 3)^2 = 13^2 \Rightarrow x^2 + x^2 + 6x + 9 = 169 \Rightarrow 2x^2 + 6x - 160 = 0 \Rightarrow x^2 + 3x - 80 = 0 \Rightarrow (x + 8)(x - 10) = 0 \Rightarrow x = 10$ (positive). Sides 10 and 13.

Part E — Linear Models (Concept)

23. Phone plan model.

Solution: (a) $C(T) = 20 + 0.08T$. (b) Slope 0.08: cost increases \$0.08 per text (marginal cost).
(c) y-intercept 20: base monthly fee with 0 texts.